My Explorations in the Collatz-Forest and Some of My Findings

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SOME ICE-BREAKER PLEASANTRIES

we may not be in an integral domain;

but, i can assure you, there are

no irrationals; no radicals; no trans; no imaginaries; nothing complex;

not even any negatives;

only positives, that too, with some modular ideals.

HISTORICAL BACKGROUND

Lothar Collatz (1910-1990); Bryan Thwaites (1923-); Stanisław Ulam (1909-1984);
Helmut Hasse (1898-1979); Syracuse University, NY; Shizuo Kakutani (1911-2004);

HISTORICAL BAGGAGE

Jeffrey C. Lagarias refers to the statement by Paul Erdos,
one of the most revered Mathematician and Number Theory Expert, who said that -
"Mathematics is not yet ready for such problems" – "Hopeless, Absolutely Hopeless" –
about the Collatz Conjecture.

Richard K. Guy has been somewhat pessimistic in advising researchers -
"Don't Try to Solve These Problems".

Jeffrey C. Lagarias gives an elaborate overview of the Collatz Problem,
in his book "The Ultimate Challenge: The 3x+1 Problem";
and also an extensive two-part annotated bibliography.
"It is hard to resist exploring its structure. We should not exclude it from the mathematical universe just because we are unhappy with its difficulty. It is a fascinating and addictive problem".

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The Collatz (3x+1) System is indeed a highly 'chaotic' system
defined or generated with one of the simplest deterministic well-defined equations,
that too, just two in number.
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Collatz Function Collatz (3x+1) Sequence Collatz Conjecture

Start with any given positive integer x. If x is even then, divide by 2 {x ⇐ x/2}; else(if x is odd) then, multiply by 3 and add 1 {x ⇐ (3x+1)} so as to get a resultant that is even. Now continue the process.

My way of defining the Collatz Function:

if (x≡0(mod2)) then S(x) ≔ [x]D ≔ (x/2);
elseif (x≡1(mod2)) then S(x) ≔ [x]U ≔ (3x+1);

The sequence thus generated, starting from any given positive integer, is the Collatz Sequence corresponding to that given positive integer.

The Collatz Conjecture states that the Collatz Sequence will eventually converge to the trivial-cycle; that is, {(1⇐2⇐4)}; starting from any positive integer.

Collatz Sequence starting from 27

27, 82, **41**, 124, 62, **31**, 94, **47**, 142, **71**, 214, **107**, 322, **161**, 484, 242, **121**, 364, 182, **91**, 274, **137**, 412, 206, **103**, 310, **155**, 466, **233**, 700, 350, **175**, 526, **263**, 790, **395**, 1186, **593**, 1780, 890, **445**, 1336, 668, 334, **167**, 502, **251**, 754, **377**, 1132, 566, **283**, 850, **425**, 1276, 638, **319**, 958, **479**, 1438, **719**, 2158, **1079**, 3238, **1619**, 4858, **2429**, 7288, 3644, 1822, **911**, 2734, **1367**, 4102, **2051**, 6154, **3077**, 9232, 4616, 2308, 1154, **577**, 1732, 866, **433**, 1300, 650, **325**, 976, 488, 244, 122, **61**, 184, 92, 46, **23**, 70, **35**, 106, **53**, **160**, 80, 40, 20, 10, **5**, 16, 8, 4, 2, **1**

27, 82, **41**, 124, **31**, 94, **47**, 142, **71**, 214, **107**, 322, **161**, 484, **121**, 364, **91**, 274, **137**, 412, **103**, 310, **155**, 466, **233**, 700, **175**, 526, **263**, 790, **395**, 1186, **593**, 1780, **445**, 1336, **167**, 502, **251**, 754, **377**, 1132, **283**, 850, **425**, 1276, **319**, 958, **479**, 1438, **719**, 2158, **1079**, 3238, **1619**, 4858, **2429**, 7288, **911**, 2734, **1367**, 4102, **2051**, 6154, **3077**, **9232**, **577**, 1732, 866, **433**, 1300, 650, **325**, 976, 488, 244, 122, **61**, 184, 92, 46, **23**, 70, **35**, 106, **53**, **160**, **5**, 16, **1**

27; 41; 31; 47; 71; 107; 161; 121; 91; 137; 103; 155; 233; 175; 263; 395; 593; 445; 167; 251; 377; 283; 425; 319; 479; 719; 1079; 1619; 2429; 911; 1367; 2051; 3077; 577; 433; 325; 61; 23; 35; 53; 5; 1

Compact Collatz (even-to-odd, odd-to-even) Function

if $x \equiv 0 \pmod 2$ then, $T(x) \coloneqq [x]D^{\#} \coloneqq [x]D^{*} \coloneqq [x]D^{p} = (x/2^{p})$; for some valid $p \geq 1$;
so as to get a resultant that is odd, $x \not\equiv 0 \pmod 2$;
elseIf $x \not\equiv 0 \pmod 2$ then, $T(x) \coloneqq [x]U^{\#} \coloneqq [x]U \coloneqq (3x+1)$; which is even, $x \equiv 0 \pmod 2$.

The compact Collatz sequence is an alternating sequence of odd and even numbers.

Condensed Compact (odd-to-odd) Collatz Function

Start with any positive odd integer $x>1$; $x \not\equiv 0 \pmod 2$; apply the operator $(U^{\#}D^{\#})$;
$x \Leftarrow [x](U^{\#}D^{\#})$; so that the resultant is odd, $x \not\equiv 0 \pmod 2$; now continue the process.

THE COLLATZ-THWAITES-ULAM-HASSE-SYRACUSE-KAKUTANI (CTUHSK) SEQUENCE

Modular Periodicity

[b]$_3$,[y]$_6$ =	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**= [b]$_3$,[y]$_6$**
j=9; w=18 **j=9; v=17**	**(1,3)**87381	218453**(0,5)**	**(1,5)**611669	480597**(2,3)**	**(2,1)**1135957	742741**(1,1)**	**j=9; w=18** **j=9; v=17**
j=8; w=16 **j=8; v=15**	**(1,5)**21845	54613**(0,1)**	**(1,1)**152917	120149**(2,5)**	**(0,3)**283989	185685**(2,3)**	**j=8; w=16** **j=8; v=15**
j=7; w=14 **j=7; v=13**	**(1,1)**5461	13653**(1,3)**	**(2,3)**38229	30037**(2,1)**	**(0,5)**70997	46421**(2,5)**	**j=7; w=14** **j=7; v=13**
j=6; w=12 **j=6; v=11**	**(2,3)**1365	3413**(1,5)**	**(2,5)**9557	1251**(0,3)**	**(0,1)**17749	11605**(2,1)**	**j=6; w=12** **j=6; v=11**
j=5; w=10 **j=5; v= 9**	**(2,5)**341	853**(1,1)**	**(2,1)**2389	1877**(0,5)**	**(1,3)**4437	2901**(0,3)**	**j=5; w=10** **j=5; v= 9**
j=4; w= 8 **j=4; v= 7**	**(2,1)**85	213**(2,3)**	**(0,3)**597	469**(0,1)**	**(1,5)**1109	725**(0,5)**	**j=4; w= 8** **j=4; v= 7**
j=3; w= 6 **j=3; v= 5**	**(0,3)**21	53**(2,5)**	**(0,5)**149	117**(1,3)**	**(1,1)**277	181**(0,1)**	**j=3; w= 6** **j=3; v= 5**
j=2; w= 4 **j=2; v= 3**	**(0,5)**5	13**(2,1)**	**(0,1)**37	29**(1,5)**	**(2,3)**69	45**(1,3)**	**j=2; w= 4** **j=2; v= 3**
j=1; w= 2 **j=1; v= 1**	**(0,1)**1	3**(0,3)**	**(1,3)**9	7**(1,1)**	**(2,5)**17	11**(1,5)**	**j=1; w= 2** **j=1; v= 1**
[b]$_3$,[y]$_6$ =	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**= [b]$_3$,[y]$_6$**

[b]$_3$,[y]$_6$ =	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**= [b]$_3$,[y]$_6$**
j=1; w= 2	**(0,1)**	3	**(1,3)**	7	**(2,5)**	11	**(1,1)**	15	**(2,3)**	19	**(0,5)**	23	**(2,1)**	27	**(0,3)**	31	**(1,5)**	35	**j=1; w= 2**
j=1; v= 1	1	**(0,3)**	9	**(1,1)**	17	**(1,5)**	**25**	**(2,3)**	**33**	**(0,1)**	**41**	**(0,5)**	**49**	**(1,3)**	**57**	**(2,1)**	**65**	**(2,5)**	**j=1; v= 1**
[b]$_3$,[y]$_6$ =	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**(0,1)**	**(0,5)**	**(1,1)**	**(1,5)**	**(2,1)**	**(2,5)**	**= [b]$_3$,[y]$_6$**

Condensed compact inverse CTUHSK odd-to-odd Function

input $x \coloneqq 6a+x_6$;
with $0 \leq a$; $a_3 \coloneqq a(\text{mod}3) \in \{1,2,0\}$; $x_3 \coloneqq x(\text{mod}3) \in \{1,2\}$ and $x_6 \coloneqq x(\text{mod}6) \in \{1,5\}$;

binary-exponent-value $k \coloneqq 2j - (x_6 - x_3)/3$;
for the j^{th} jump; $1 \leq j$; $j_3 \coloneqq j(\text{mod}3) \in \{1,2,0\}$; k is the binary-exponent-value;

output $y \coloneqq 6b+y_6$;
with $0 \leq b$; $b_3 \coloneqq b(\text{mod}3) \in \{1,2,0\}$; $y_3 \coloneqq y(\text{mod}3) \in \{1,2,0\}$ and $y_6 \coloneqq y(\text{mod}6) \in \{1,5,3\}$;

Fruits of the CTUHSK Arborescence

Odd multiples of 3 reached through condensed compact inverse Collatz function applied on

BEL(6a+1); $0 \leq a$; $(j_3-a_3)=3c$;
hanging from the rung defined by the binary-exponent-value $k \coloneqq (2j) \coloneqq 2(3c+a_3)$; $0 \leq i$;

and

BEL(6a+5); $0 \leq a$; $(j_3+a_3-1)=3c$;
hanging from the rung defined by the binary-exponent-value $k \coloneqq (2j-1) \coloneqq 1+2(3c-a_3)$; $0 \leq i$;

Recommended Reading

[§1]. Wikipedia Page – https://en.wikipedia.org/wiki/Collatz_conjecture

[§2]. Richard K Guy; "Don't Try to Solve These Problems";
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[§5]. Jeffrey C Lagarias; "The 3x + 1 Problem: An Overview";
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[§6]. Jeffrey C Lagarias; "The Ultimate Challenge: The 3x+1 Problem";
American Mathematical Society; 2010.

[§7]. Keshava Prasad Halemane; "Collatz-Conjecture Proved and Halemane-Conjecture Proposed";
https://doi.org/10.31224/6063; 2025.

[§8]. Keshava Prasad Halemane; "Halemane's Theory on the Collatz (3x+1) System";
https://doi.org/10.31224/7569; 2026.

SOME SIMPLER CREATURES

Halemane's (1.x+1)/2 System (even-to-odd, odd-to-even) Form

Start with any given positive integer x;
If $x \equiv 0 \pmod 2$ then, divide by 2 repeatedly, until the resultant $x \equiv 1 \pmod 2$;
ElseIf $x \equiv 1 \pmod 2$ then, add 1 to get a resultant $x \equiv 0 \pmod 2$;
Now continue the process.

if $x \equiv 0 \pmod 2$ then, $T(x) \coloneqq [x]D^{\#} \coloneqq [x]D^{*} \coloneqq [x]D^{p} = (x/2^{p})$;
elseIf $x \equiv 1 \pmod 2$ then, $T(x) \coloneqq (1.x+1)$; so that the resultant is even.

Inverse of (1.x+1)/2 System - Condensed Compact (odd-to-odd) Form

if $y \equiv 1 \pmod 2$ then $\{P(y)\} \coloneqq \{P_1(y, u) \coloneqq (y.2^{u} - 1)\}$; $1 \leq u$.

Halemane's (2.x±1)/3 System

Start with any given positive integer x;
If $x \equiv 0 \pmod 3$ then, divide by 3 repeatedly until the resultant $x \not\equiv 0 \pmod 3$;
ElseIf $x \equiv 1 \pmod 3$ then, multiply by 2 and add 1 to get a resultant $x \equiv 0 \pmod 3$;
ElseIf $x \equiv 2 \pmod 3$ then, multiply by 2 and subtract 1 to get a resultant $x \equiv 0 \pmod 3$;
Now continue the process.

Inverse of (2.x±1)/3 System

if $\{(y \equiv 1 \pmod 6)) \vee ((y \equiv 5 \pmod 6))\}$ then $\{x\} \coloneqq \{P(y)\} \coloneqq (y.3^u \pm 1)/2\}$; with $1 \leq u$.

Halemane's (3.x±1)/4 System

Start with any given positive integer $x>1$; if ($x=1$) STOP;
elseIf $x\equiv 0 \pmod 4$ then, divide by 4 repeatedly until the resultant $x\not\equiv 0 \pmod 4$;
ElseIf $x\equiv 2 \pmod 4$ then, divide by 2 to get a resultant $x\not\equiv 2 \pmod 4$;
ElseIf $\{x\equiv 1 \pmod 4\}$ then, multiply by 3 and add 1 to get a resultant $\{x\equiv 0 \pmod 4\}$;
ElseIf $\{x\equiv 3 \pmod 4\}$ then, multiply by 3 and subtract 1 to get a resultant $\{x\equiv 0 \pmod 4\}$;
Now continue the process.

Halemane's (3.x±1)/4 Function

if ($x=1$) STOP;
elseIf $x\equiv 0 \pmod 4$ then, $S(x) \coloneqq (x/4)$; elseIf $x\equiv 2 \pmod 4$ then, $S(x) \coloneqq (x/2)$;
elseIf $\{x\equiv 1 \pmod 4)\}\}$ then, $S(x) \coloneqq (3.x+1)$;
elseIf $\{x\equiv 3 \pmod 4)\}\}$ then, $S(x) \coloneqq (3.x-1)$;

	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	0	1	1	1
$[1]_4$	1	0	0	0
$[2]_4$	0	1	0	1
$[3]_4$	1	0	0	0

Halemane's (3.x+1)/4 System - as a clone of the Collatz System

Start with any given positive integer x>1; if (x=1) STOP;
elseIf $x\equiv 0 \pmod 4$ then, divide by 4 repeatedly until the resultant $x\not\equiv 0 \pmod 4$;
ElseIf $x\equiv 2 \pmod 4$ then, multiply by 2 to get a resultant $x\equiv 0 \pmod 4$;
ElseIf $\{x\equiv 1 \pmod 4\}\vee\{x\equiv 3 \pmod 4\}$ then, multiply by 3 and add 1 to get a resultant $\{x\equiv 0 \pmod 4\}\vee\{x\equiv 2 \pmod 4\}$; Now continue the process.

Halemane's (3.x+1)/4 Function

if (x=1) STOP;
elseIf $x\equiv 0 \pmod 4$ then, $S(x) \coloneqq (x/4)$; elseIf $x\equiv 2 \pmod 4$ then, $S(x) \coloneqq (2x)$;
elseIf $\{x\equiv 1 \pmod 4\}\vee\{x\equiv 3 \pmod 4\}$ then, $S(x) \coloneqq (3.x+1)$;

	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	0	1	1	1
$[1]_4$	1	0	0	0
$[2]_4$	1	0	0	0
$[3]_4$	0	0	1	0

Halemane's (3.x+1)/4 System - modified

Start with any given positive integer x>1; if (x=1) STOP;
elseIf $x\equiv0 \pmod 4$ then, divide by 4 repeatedly until the resultant $x\not\equiv0 \pmod 4$;
ElseIf $x\equiv2 \pmod 4$ then, divide by 2 to get a resultant $x\not\equiv2 \pmod 4$;
ElseIf $\{x\equiv1 \pmod 4\}\vee\{x\equiv3 \pmod 4\}$ then, multiply by 3 and add 1 to get a resultant $\{x\equiv0 \pmod 4\}\vee\{x\equiv2 \pmod 4\}$; Now continue the process.

Halemane's (3.x+1)/4 Function

if (x=1) STOP;
elseIf $x\equiv0 \pmod 4$ then, $S(x) \coloneqq (x/4)$; elseIf $x\equiv2 \pmod 4$ then, $S(x) \coloneqq (x/2)$;
elseIf $\{x\equiv1 \pmod 4\}\vee\{x\equiv3 \pmod 4\}$ then, $S(x) \coloneqq (3.x+1)$;

	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	0	1	1	1
$[1]_4$	1	0	0	0
$[2]_4$	0	1	0	1
$[3]_4$	0	0	1	0

EXPLORATORY-ROUGH-WORK

[6k-1] = [4k-1](UD);

if $[m](UD)^k = n$; then $(n+1)/(m+1) = (3/2)^k$;
or equivalently, $(n+1).2^k = (m+1).3^k$;
or equivalently, $[2.2^k -1](UD)^k = [2.3^k -1]$;

If $x \equiv 0 (\mathrm{mod} 2)$; let $x = (a.2^p -1).2^q$ for some $a \equiv 1 (\mathrm{mod} 2)$ and $0<p$ and $0<q$;
then, $x \Leftarrow [x](D)^q = [x]/2^q = (a.2^p -1)$; which is an odd number;
ElseIf $x \equiv 1 (\mathrm{mod} 2)$ and if $x = (a.2.2^p -1)$ for some $a \equiv 1 (\mathrm{mod} 2)$ and $0 \leq p$;
then, $x \Leftarrow [x](UD)^q = (a.2.3^p -1$; which is an odd number.

[6k+1] = [8k+1](UDD) ;

if $[m](UDD)^k = n$; then $(n-1)/(m-1) = (3/4)^k$;
or equivalently, $(n-1).4^k = (m-1).3^k$;
or equivalently, $[4^k +1](UDD)^k = [3^k +1]$;

If $x \equiv 0 (\mathrm{mod} 2)$; let $x = (a.2^p -1).4^q$ for some $a \equiv 1 (\mathrm{mod} 2)$ and $0<p$ and $0<q$;
then, $x \Leftarrow [x](DD)^q = [x]/4^q = (a.2^p -1)$; which is an odd number;
ElseIf $x \equiv 1 (\mathrm{mod} 2)$ and if $x = (a.4.4^q +1)$ for some $a \equiv 1 (\mathrm{mod} 2)$ and $0 \leq q$;
then, $x \Leftarrow [x](UDD)^q = (a.4.3^q -1)$; which is an odd number.

$$z = (6b + 1) \Leftarrow (8b + 1) = y = (6a - 1) \Leftarrow (4a - 1) = x$$

$$4(z - 1) = 3(y - 1) \text{ and } 2(y + 1) = 3(x + 1)$$

$$8(z - 1) = 6(y - 1) \text{ and } 6(y + 1) = 9(x + 1)$$

$$2(4z - 1) = 6(y) = 3(3x + 1)$$

z	z⇐y	y	!=!	y	y⇐x	x
1	z⇐y(UDD)D^2	5		5	y⇐x(UD)	3
5	z⇐y(UDD)D	13		5	y⇐x$(UD)^4(D)^3$	15
13	z⇐y(UDD)	17		17	y⇐x(UD)	11
7	z⇐y(UDD)	9		11	y⇐x(UD)	7
11	z⇐y(UDD)D	29		29	y⇐x(UD)	19
19	z⇐y(UDD)	25		35	y⇐x(UD)	23
25	z⇐y(UDD)	33		47	y⇐x(UD)	31
31	z⇐y(UDD)	41		41	y⇐x(UD)	27
z	z⇐y	y		y	y⇐x	x

1⇐5$(UDD)^2$; 5⇐13(UDD)D; 13⇐17(UDD); 17⇐11(UD); 11⇐7(UD); 7⇐9(UDD);
5⇐3(UD); 5⇐15$(UD)^4(D)^3$; 1⇐21(UDD)$(D)^4$;